

Implications of Healthcare Technology

- Population increase and issues of chronic diseases
- IoT technology in healthcare
- Rapid growth

A few examples:

Smart thermometers

Connected inhalers

Smart watches

Fitness trackers (FitBits)

ECG monitors

Blood pressure monitors





- More convenient and cheaper in many ways:
- Wireless medical devices can greatly reduce the **financial burden** of public healthcare
 - Telehealth or self-service medical treatment is realized **through informatization**
 - Medical information sharing can help healthcare platforms to be developed **equally across the world**
 - It is conducive to the **modernization of healthcare services** and the improvement of healthcare standards



Ethical issues of smart healthcare

- (refer to the article given)

Ageing, Old age and Technoscience

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- The relationship between ageing, old age and technology is quite critical... (either incapable or resistant), but technology helps them
 - Mostly in maintaining their physical autonomy and reduce their care-dependency (however, technology is not limited to technological tools, rather beyond)
 - Four postulates of ageing:
 - **Ageing is universal:** a phenomenon associated with the process of ageing must occur in different degrees in all individuals of a species.
 - **Ageing must be intrinsic:** the causes that are the origin of ageing must be endogenous; they must not depend on extrinsic factors
 - **Ageing must be progressive:** changes that lead to ageing must occur progressively throughout the life span (they must also occur in young individuals, albeit in a small proportion)
 - **Ageing must be deleterious:** i.e., a phenomenon associated with ageing will only be considered as part of the ageing process if it is 'bad' for the individual



- There can be broadly the following reasons:
 - First, older people are not a homogeneous group, but a very diverse ...making it difficult to generalize the '**needs of older people**'
 - Secondly, ageing is not only a biomedical bodily phenomenon, rather it has psychological, social, political, and economical dimensions...the social image of older people define the technological intervention
 - Third, ageing is not an isolated phase of an individual life, it rather is part of the life course...people age within their social and historical contexts...
 - Fourth, our understanding of ageing is shaped by powerful biomedical, economical and welfare state discourses...ageing is related to being productive...if technology for the aged is supporting such economic view on ageing or emphasizing on making their life active
- Technology for older people is affected not only by different perspectives of ageing but also the participation of older people in the process of technology development... (their experience, social context, adaptation etc...developing the ergonomics)



A few examples of tech devices for older people

- Robotics for (physical, social, cognitive) assistance
 - Telecare to allow care at a distance
 - Personal alarm systems for safety
 - Health apps to coach healthy behavior
 - Medicine dispensers for therapeutic adherence etc.
- These technologies often have a digital or ICT component in them, aiming to improve communication or stimulate daily activity and health
 - **Paro**: A robot in the form of a baby seal
 - This was initially designed to stimulate **group activities** among older people living in old age homes/care homes
 - Increasingly, Paro is used in a **therapeutic manner** for those elderly people who have problems with emotional contact and sleeping
 - <https://www.youtube.com/watch?v=oJq5PQZHU-I>